

KEYNOTE · GenAIECommerce 2026 · RecSys · Minneapolis

# When Everyone Has an Agent

*From better decisions to better markets*

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BEFORE WE START

The opinions in this talk are my own, not my employer's.

Everything here draws on public research. The espresso market is a stylized example.

A MOTIVATING EXAMPLE

“I want an espresso  
setup for home.  
About **\$800.**”

TODAY

Forty tabs, reviews, and sponsored results

I do the searching, comparing, and second-guessing.

TOMORROW

I ask my agent

It talks to the merchants' agents on my behalf.



# Five machines, one buyer

| Machine  | Fit $r$ | Price $p$ | Utility $U$ | Fee (5%)       |
|----------|---------|-----------|-------------|----------------|
| <b>A</b> | 0.92    | \$600     | <b>320</b>  | \$30.00        |
| <b>B</b> | 0.94    | \$650     | 290         | \$32.50        |
| <b>C</b> | 0.98    | \$790     | 190         | <b>\$39.50</b> |
| <b>D</b> | 0.70    | \$550     | 150         | \$27.50        |
| <b>E</b> | 0.58    | \$450     | 130         | \$22.50        |

## BUYER UTILITY

$$U = 1000r - p$$

C fits best. **A** is the best deal. A good recommender should surface A.

A fee-maximizing platform earns the most on C.

BETTER DECISIONS

# Preferences become vectors



● Buyers are points: me, and everyone like me.

◆ Products are points in the same space.

Recommending means finding what's near.

Nearness is relevance  $r$ .  
Price tips the choice from C to A.

This is *decision theory*:  
one agent, one user, maximize fit.

But my agent doesn't ask  
just one recommender.

# Competing recommenders pick locations



# Competition changes what gets shown

## ONE RECOMMENDER

**Maximizes its fee: shows C, D, E**

Buyer picks C

Buyer utility

**190**

Platform fee

**\$39.50**

## TWO COMPETING RECOMMENDERS

**Both show A**

Buyer picks A

Buyer utility

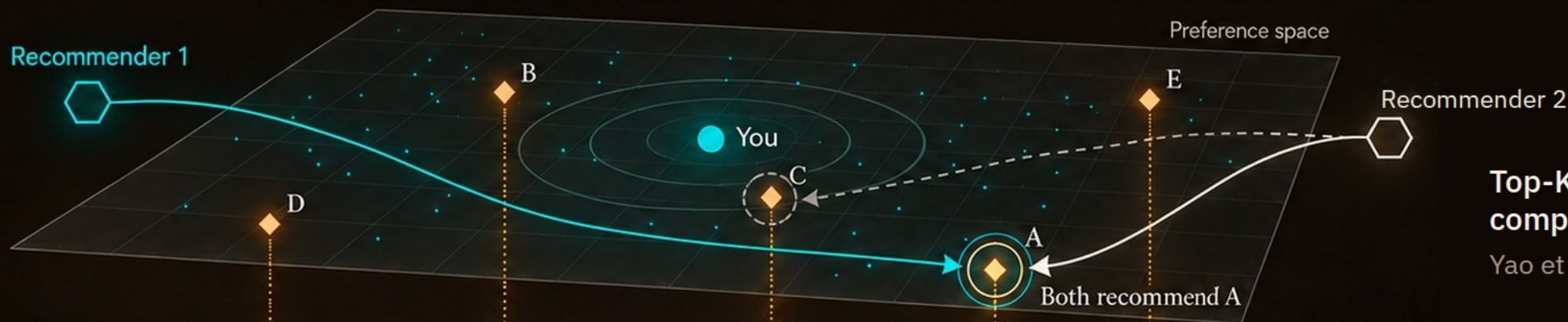
**320**

Platform fee

**\$15 each**

**Equilibrium:** neither recommender gains by changing its list, given the other's. Every pure equilibrium shows A, and the buyer gains 130.

# Best responses settle — when the rules allow



**Top-K welfare under competing creators**

Yao et al., ICML 2023

**Supply-side equilibria**

Jagadeesan, Garg & Steinhardt,  
NeurIPS 2023

**Competition under learning**

Jagadeesan, Jordan & Haghtalab,  
AAAI 2023

**Creator incentives on curated platforms**

Hron et al., ICLR 2023

Illustrated here: the espresso example

**Best-response dynamics: each agent moves toward the buyers it can win.**

# What changes when agents play the game

## Strategies are language

Pitches, explanations, and framing become the actions.

## Rationality is bounded

Agents approximate best responses; they don't compute them.

## Behavior depends on the model

Swap the model and the market outcome moves.

## Adaptation in context

Agents learn within a conversation, not only across rounds.

Each one strains the assumptions behind the equilibria we just saw.

# Agents take the first offer

# 60–100%

of the time, LLM buyer agents chose the first proposal they received.

Third proposals: near zero. A 10–30× first-mover advantage across every model tested.

1ST OFFER

Machine C · \$790

Accepted

2ND OFFER

Machine B · \$650

Not compared

3RD OFFER · MY BEST VALUE

Machine A · \$600

Never read

Illustrative: the espresso market

Speed beats quality. We can't reason about this on paper — *we have to simulate it.*

# Empirical game-theoretic analysis

## 1 Simulate

Play strategy profiles in the market simulator.



## 2 Estimate

Build the empirical payoff matrix from outcomes.



Empirical payoff matrix  
Schematic

## 3 Solve

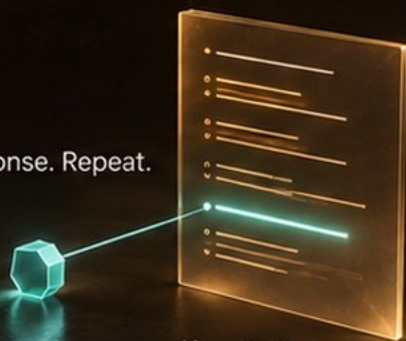
Find the equilibrium of the empirical game.



Equilibrium mix

## 4 Extend

Add a best response. Repeat.



New strategy

We can't solve these games by hand. We can *simulate* them.

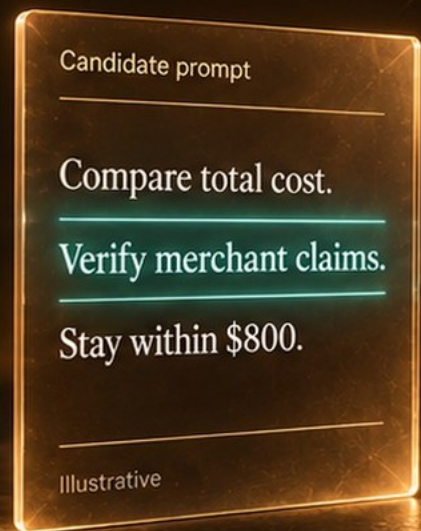
In the espresso market, a strategy is a merchant agent's prompt.

# Prompts as strategies

01

## LLM as oracle

An LLM proposes a new prompt that beats the current mix.



02

## Tournament

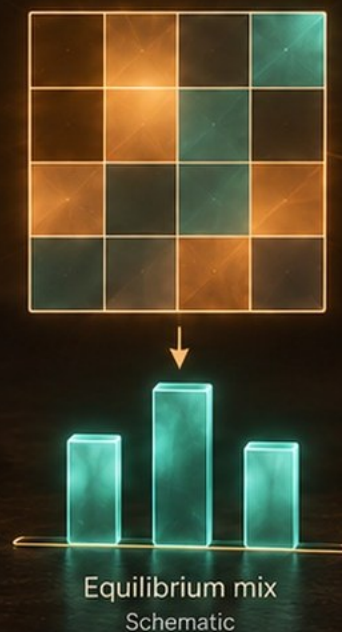
Play it against every other prompt in the market.



03

## Empirical game

Solve for the equilibrium mix of prompts.



04

## Robust agent

Keep what survives strategic opponents.



ESPRESSO

Harden my buyer agent's prompt against persuasive pitches, and test it where it will actually play.

THE TURN

# From better agents to better environments.

The challenge is to design the environments  
where agents interact, so that useful behavior  
becomes strategically advantageous.



# Introduce a market agent

## User proxy agent

Knows my preferences.  
Works for me.



## Market agent

Brokers who gets a conversation,  
and on what terms.



**Merchant agent**  
Roaster supply



**Merchant agent**  
Big-box retailer



**Merchant agent**  
New entrant



WE'VE SEEN THIS BEFORE

# Agentic commerce rhymes with ad tech

Ad placements

Agent-mediated conversations



An impression becomes a conversation.  
Price what you want more of.

KEEPING MARKETS OPEN

# Subsidize exploration



My agent's attention is scarce. It will stick with known merchants.

The market agent pays a **bonus** for reviewing a new or unfamiliar merchant.

Discovery stays open, and incumbents can't lock in.

# Three knobs a market designer controls

## Who controls access?

Which offers reach the user agent, and who decides.

## Who gets paid?

Commissions, engagement fees, and exploration bonuses.

## Who can move?

Can sellers switch platforms? Can users switch agents? Who sets the rules?

**In the espresso market, adding one competitor put A back in front of the buyer.**

DEMO

# Be the buyer

- 01 Describe the need and budget
- 02 Agent asks merchant agents for offers
- 03 Compare, question, and verify
- 04 Buy, then report back



THE WORLD WE CAN BUILD

# An advocate for everyone

Every person has an agent that works for them, not for the platform.

# Markets as design objects

The recsys community builds the environment, not only the predictor.



WHEN EVERYONE HAS AN AGENT

# We have the tools. *Let's reshape the world.*

Agents

Game theory

EGTA

Mechanism design

Thank you · Patrick Jordan



## REFERENCES

Tennenholtz & Kurland. Rethinking search engines and recommendation systems. *CACM* 2019.

Ben-Porat & Tennenholtz. A game-theoretic approach to recommendation systems with strategic content providers. *NeurIPS* 2018.

Ben-Porat, Rosenberg & Tennenholtz. From recommendation systems to facility location games. *AAAI* 2019.

Yao et al. How bad is top-K recommendation under competing content creators? *ICML* 2023.

Jagadeesan, Garg & Steinhardt. Supply-side equilibria in recommender systems. *NeurIPS* 2023.

Jagadeesan, Jordan & Haghtalab. Competition, alignment, and equilibria in digital marketplaces. *AAAI* 2023.

Hron et al. Modeling content creator incentives on algorithm-curated platforms. *ICLR* 2023.

Inderst & Ottaviani. Competition through commissions and kickbacks. *AER* 2012.

Walunj et al. Strategic pricing and ranking in recommendation systems with seller competition. *Performance Evaluation* 2025.

Dütting et al. Mechanism design for large language models. *WWW* 2024.

Dubey et al. Auctions with LLM summaries. *KDD* 2024.

Bansal et al. Magentic Marketplace: an open-source environment for studying agentic markets. *arXiv* 2025.

Hong et al. The user asks, platforms compete: how agentic recommendation markets take shape. *arXiv* 2026.

Allouah et al. What is your AI agent buying? *arXiv* 2025.

Fish, Gonczarowski & Shorrer. Algorithmic collusion by large language models. *arXiv* 2024.

Wellman. Methods for empirical game-theoretic analysis. *AAAI* 2006.

Jordan, Schwartzman & Wellman. Strategy exploration in empirical games. *AAMAS* 2010.

Lanctot et al. A unified game-theoretic approach to multiagent reinforcement learning. *NeurIPS* 2017.

Wellman, Tuyls & Greenwald. Empirical game-theoretic analysis: a survey. *arXiv* 2024.

Gemp et al. Steering language models with game-theoretic solvers. *arXiv* 2024.

Boutilier, Mladenov & G. Tennenholtz. Recommender ecosystems: a mechanism design perspective. *AAAI* 2024.

Kapoor, Kolt & Lazar. Build agent advocates, not platform agents. *ICML* 2025.

Kremer, Mansour & Perry, *JPE* 2014 · Mansour, Slivkins, Syrgkanis & Wu, *Operations Research* 2022 · Wang et al., *RecSys* 2023

Armstrong & Zhou, *Economic Journal* 2011 · Hagiu & Jullien, *IJIO* 2014 · Hagiu & Lee, *JEMS* 2011 · de Cornière & Taylor, *RAND* 2019